

we assume that treatment is binary, but our results apply to any ordered treatment, as we show in Section ?? of the Web Appendix. Then, for every $(i, g, t) \in \{1, \dots, N_{g,t}\} \times \{1, \dots, G\} \times \{1, \dots, T\}$, let $D_{i,g,t}$ and $(Y_{i,g,t}(0), Y_{i,g,t}(1))$ respectively denote the treatment status and the potential outcomes without and with treatment of observation i in group g at period t .

The outcome of observation i in group g and period t is $Y_{i,g,t} = Y_{i,g,t}(D_{i,g,t})$. For all (g, t) , let

$$D_{g,t} = \frac{1}{N_{g,t}} \sum_{i=1}^{N_{g,t}} D_{i,g,t}, \quad Y_{g,t}(0) = \frac{1}{N_{g,t}} \sum_{i=1}^{N_{g,t}} Y_{i,g,t}(0), \quad Y_{g,t}(1) = \frac{1}{N_{g,t}} \sum_{i=1}^{N_{g,t}} Y_{i,g,t}(1), \quad \text{and} \quad Y_{g,t} = \frac{1}{N_{g,t}} \sum_{i=1}^{N_{g,t}} Y_{i,g,t}.$$

$D_{g,t}$ denotes the average treatment in group g at period t , while $Y_{g,t}(0)$, $Y_{g,t}(1)$, and $Y_{g,t}$ respectively denote the average potential outcomes without and with treatment and the average observed outcome in group g at period t .

Throughout the paper, we maintain the following assumptions.

Assumption 1 (*Balanced panel of groups*) For all $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, $N_{g,t} > 0$.

Assumption 1 requires that no group appears or disappears over time. This assumption is often satisfied. Without it, our results still hold but the notation becomes more complicated as the denominators of some of the fractions below may then be equal to zero.

Assumption 2 (*Sharp design*) For all $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$ and $i \in \{1, \dots, N_{g,t}\}$, $D_{i,g,t} = D_{g,t}$.

Assumption 2 requires that units' treatments do not vary within each (g, t) cell, a situation we refer to as a sharp design. This is for instance satisfied when the treatment is a group-level variable, for instance a county- or a state-law. This is also mechanically satisfied when $N_{g,t} = 1$. In our survey in Section 6.1, we find that almost 80% of the papers using two-way fixed effects regressions and published in the AER between 2010 and 2012 consider sharp designs. We focus on sharp designs because of their prevalence, but in Section ?? of the Web Appendix, we show that all the results in Sections 3-4 below can be extended to fuzzy designs.

Assumption 3 (*Independent groups*) The vectors $(Y_{g,t}(0), Y_{g,t}(1), D_{g,t})_{1 \leq t \leq T}$ are mutually independent.

We consider $D_{g,t}$, $Y_{g,t}(0)$, $Y_{g,t}(1)$ as random variables. For instance, aggregate random shocks may affect the average potential outcomes of group g at period t . The treatment status of group g at period t may also be random. The expectations below are taken with respect to the distribution of those random variables. Assumption 3 allows for the possibility that the treatments and potential outcomes of a group may be correlated over time, but it requires that the potential outcomes and treatments of different groups be independent.

Assumption 4 (*Strong exogeneity*) For all $(g, t) \in \{1, \dots, G\} \times \{2, \dots, T\}$,
 $E(Y_{g,t}(0) - Y_{g,t-1}(0) | D_{g,1}, \dots, D_{g,T}) = E(Y_{g,t}(0) - Y_{g,t-1}(0)).$

Assumption 4 requires that the shocks affecting a group's $Y_{g,t}(0)$ be mean independent of that group's treatment sequence. This rules out the possibility that a group gets treated because it experiences negative shocks, the so-called Ashenfelter's dip (see Ashenfelter, 1978). Assumption 4 is related to the strong exogeneity condition in panel data models, which, as is well-known, is necessary to obtain the consistency of the fixed effects estimator (see, e.g., Wooldridge, 2002).

We now define the FE regression described in the introduction.⁴

Regression 1 (*Fixed-effects regression*)

Let $\hat{\beta}_{fe}$ denote the coefficient of $D_{g,t}$ in an OLS regression of $Y_{i,g,t}$ on group fixed effects, period fixed effects, and $D_{g,t}$. Let $\beta_{fe} = E[\hat{\beta}_{fe}]$.⁵

For all g and t , let $N_{g,\cdot} = \sum_{t=1}^T N_{g,t}$ and $N_{\cdot,t} = \sum_{g=1}^G N_{g,t}$ respectively denote the total number of observations in group g and in period t . For any variable $X_{g,t}$ defined in each (g, t) cell, let $X_{g,\cdot} = \sum_{t=1}^T (N_{g,t}/N_{g,\cdot}) X_{g,t}$ denote the average value of $X_{g,t}$ in group g , let $X_{\cdot,t} = \sum_{g=1}^G (N_{g,t}/N_{\cdot,t}) X_{g,t}$ denote the average value of $X_{g,t}$ in period t , and let $X_{\cdot,\cdot} = \sum_{g,t} (N_{g,t}/N) X_{g,t}$ denote the average value of $X_{g,t}$. For instance, $D_{3,\cdot}$ and $D_{\cdot,2}$ respectively denote the average treatment in group 3 across time and in period 2 across groups, whereas $Y_{\cdot,\cdot}$ denotes the average value of the outcome across groups and time. Finally, for any variable $X_{g,t}$, we let $\mathbf{X}$ denote the vector $(X_{g,t})_{(g,t) \in \{1, \dots, G\} \times \{1, \dots, T\}}$ collecting the values of that variable in each (g, t) cell. For instance, $\mathbf{D}$ is the vector $(D_{g,t})_{(g,t) \in \{1, \dots, G\} \times \{1, \dots, T\}}$ collecting the treatments of all the (g, t) cells.

3 Two-way fixed effects regressions

3.1 A decomposition result

We study the FE regression under the following common trends assumption.

Assumption 5 (*Common trends*) For $t \geq 2$, $E(Y_{g,t}(0) - Y_{g,t-1}(0))$ does not vary across g .

Assumption 5 requires that the expectation of the outcome without treatment follow the same evolution over time in every group. When t represents birth cohorts, Assumption 5 requires that the outcome difference between consecutive cohorts be the same across groups.

⁴ Throughout the paper, we assume that $D_{g,t}$ in Regression 1 and $D_{g,t} - D_{g,t-1}$ in Regression 2 below are not collinear with the other independent variables in those regressions, so $\hat{\beta}_{fe}$ and $\hat{\beta}_{fd}$ are well-defined.

⁵ As the independent variables in Regression 1 are constant within each (g, t) cell, Regression 1 is equivalent to a (g, t) -level regression of $Y_{g,t}$ on group and period fixed effects and $D_{g,t}$, weighted by $N_{g,t}$.

Let $N_1 = \sum_{i,g,t} D_{i,g,t}$ denote the number of treated units, let

$$\Delta^{TR} = \frac{1}{N_1} \sum_{(i,g,t): D_{i,g,t}=1} [Y_{i,g,t}(1) - Y_{i,g,t}(0)]$$

denote the average treatment effect across all treated units, and let $\delta^{TR} = E[\Delta^{TR}]$ denote the expectation of that parameter, hereafter referred to as the ATT. For any $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, let

$$\Delta_{g,t} = \frac{1}{N_{g,t}} \sum_{i=1}^{N_{g,t}} [Y_{i,g,t}(1) - Y_{i,g,t}(0)]$$

denote the ATE in cell (g, t) . δ^{TR} is equal to the expectation of a weighted average of the treated cells' $\Delta_{g,t}$ s:

$$\delta^{TR} = E \left[\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} \Delta_{g,t} \right]. \quad (2)$$

Under the common trends assumption, we show that β_{fe} is also equal to the expectation of a weighted sum of the $\Delta_{g,t}$ s, with potentially some negative weights.

Let $\varepsilon_{g,t}$ denote the residual of observations in cell (g, t) in the regression of $D_{g,t}$ on group and period fixed effects:⁶

$$D_{g,t} = \alpha + \gamma_g + \lambda_t + \varepsilon_{g,t}.$$

One can show that if the regressors in Regression 1 are not collinear, the average value of $\varepsilon_{g,t}$ across all treated (g, t) cells differs from 0: $\sum_{(g,t): D_{g,t}=1} (N_{g,t}/N_1) \varepsilon_{g,t} \neq 0$. Then we let $w_{g,t}$ denote $\varepsilon_{g,t}$ divided by that average:

$$w_{g,t} = \frac{\varepsilon_{g,t}}{\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} \varepsilon_{g,t}}.$$

Theorem 1 *Suppose that Assumptions 1-5 hold. Then,*⁷

$$\beta_{fe} = E \left[\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} \Delta_{g,t} \right].$$

⁶ $\varepsilon_{g,t}$ arises from a unit-level regression, where the dependent and independent variables only vary at the (g, t) level. Therefore, all the units in the same (g, t) cell have the same value of $\varepsilon_{g,t}$.

⁷ In the proof, we show the following, stronger result:

$$E[\hat{\beta}_{fe} | \mathbf{D}] = \sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} E[\Delta_{g,t} | \mathbf{D}].$$

This result implies that in general, $\beta_{fe} \neq \delta^{TR}$, so $\hat{\beta}_{fe}$ is a biased estimator of the ATT. To illustrate this, we consider a simple example of a staggered adoption design with two groups and three periods, and where the treatments are non-stochastic: group 1 is untreated at periods 1 and 2 and treated at period 3, while group 2 is untreated at period 1 and treated both at periods 2 and 3.⁸ We also assume that $N_{g,t}/N_{g,t-1}$ does not vary across g : all groups experience the same growth of their number of observations from $t - 1$ to t , a requirement that is for instance satisfied when the data is a balanced panel. Then, one can show that

$$\varepsilon_{g,t} = D_{g,t} - D_{g,\cdot} - D_{\cdot,t} + D_{\cdot,\cdot},$$

thus implying that

$$\begin{aligned}\varepsilon_{1,3} &= 1 - 1/3 - 1 + 1/2 = 1/6, \\ \varepsilon_{2,2} &= 1 - 2/3 - 1/2 + 1/2 = 1/3, \\ \varepsilon_{2,3} &= 1 - 2/3 - 1 + 1/2 = -1/6.\end{aligned}$$

The residual is negative in group 2 and period 3, because the regression predicts a treatment probability larger than one in that cell, a classic extrapolation problem with linear regressions. Then, under the common trends assumption, it follows from Theorem 1 and the fact that the treatments are non-stochastic that

$$\beta_{fe} = 1/2E[\Delta_{1,3}] + E[\Delta_{2,2}] - 1/2E[\Delta_{2,3}].$$

β_{fe} is equal to a weighted sum of the ATEs in group 1 at period 3, group 2 at period 2, and group 2 at period 3, the three treated (g, t) cells. However, the weight assigned to each ATE differs from $1/3$, the proportion that each cell accounts for in the population of treated observations. Therefore, β_{fe} is not equal to δ^{TR} . Perhaps more worryingly, not all the weights are positive: the weight assigned to the ATE in group 2 period 3 is strictly negative. Consequently, β_{fe} may be a very misleading measure of the treatment effect. Assume for instance that $E[\Delta_{1,3}] = E[\Delta_{2,2}] = 1$ and $E[\Delta_{2,3}] = 4$. At the period when they start receiving the treatment, both groups experience a modest positive ATE. But this effect builds over time and in period 3, one period after it has started receiving the treatment, group 2 now experiences a large ATE. Then,

$$\beta_{fe} = 1/2 \times 1 + 1 - 1/2 \times 4 = -1/2.$$

β_{fe} is strictly negative, while $E[\Delta_{1,3}]$, $E[\Delta_{2,2}]$, and $E[\Delta_{2,3}]$ are all positive. More generally, the negative weights are an issue if the $E[\Delta_{g,t}]$ s are heterogeneous, across groups or over time.⁹ If $E[\Delta_{1,3}] = E[\Delta_{2,2}] = E[\Delta_{2,3}] = 1$, then $\beta_{fe} = 1 = \delta^{TR}$.

⁸ A similar example appears in Borusyak and Jaravel (2017).

⁹ On the other hand, β_{fe} does not rule out heterogeneous treatment effects within (g, t) cells, as it is identified by variations across (g, t) cells, and does not leverage any within-cell variation.